Government College University Faisalabad
Department of Information Technology

Final Year Project - Proposal

Class:	BS(IT)	[✓] Regular Morning [ ] Shifted Morning [ ] Bridging A	[ ] Regular Evening [ ] Shifted Evening [ ] Bridging B

Particulars of the students (1 or 2 students per Group):

Sr	Roll No	Registration#	Name in Full (Use Block Letters)	Email	Signature
1	222099	2022-GCUF-02674	ALISHBAH RASHEED	buttalish661@gmail.com	
2	222134	2022-GCUF-02709	AHMAD HASEEB	ah4170537@gmail.com	

Project Title: Career Path Prediction & Skill Gap Analysis System

Domain Area
- ☐ Web Application
- ☑ Mobile Application
- ☑ Artificial Intelligence
- ☐ IoT / Embedded Systems
- ☐ Data Science
- ☐ Cyber Security
- ☐ Other: _____

Brief Description of Project (Maximum 200 words):

This project aims to develop an intelligent career guidance system for university students, specifically aligned with the **Pakistan tech industry**. Many students are confused about career selection and lack awareness of industry-required skills. As a result, there is a mismatch between students' abilities and job market demands.

The proposed system will analyze a student's academic background, current technical skills, and interests, and then recommend **suitable career paths** such as web development, data analysis, software engineering, etc. It will also identify **skill gaps** by comparing student skills with real job requirements collected from Pakistani job portals like Rozee.pk and LinkedIn.

The system will help students make informed career decisions, improve employability, and better prepare for Pakistan's technology job market.

Major Components/Modules:

Module 1: Student Profile & Skill Analysis

This module collects and analyzes student information such as education, skills, interests, and experience to understand the student's current profile.

Module 2: Career Prediction & Skill Gap Analysis

This module suggests suitable career options and identifies missing skills by comparing the student's profile with current job market requirements.

Supervisor:

Name: Dr. M. Yamin **Signature:** M. Yamin

Module 3: Career Roadmap & Industry Guidance

This module provides a step-by-step learning and career roadmap, guiding students from their current skills toward internships and better job opportunities.

Tools and Technologies to be used:

Frontend

- React native
- HTML, CSS, JavaScript
- Bootstrap / Tailwind CSS

Backend

- Node.js with Express **or** Python (Flask/Django)
- RESTful APIs

Database

- MongoDB **or** MySQL

AI / Data Processing

- Python
- Scikit-learn (for prediction and analysis)

Supervisor:

Name: .....Dr. M. Yamu..... **Signature:** .....M. Yamu..... 22-12-25